

**Specifications for:
Electronic Message Sign Boards
For Delivery to
Hawaii Volcanoes National Park
1 Crater Rim Drive HNP, HI 96718**

OVERVIEW

Hawaii Volcanoes National Park (HAVO) intends to purchase commercially available Electronic Message Sign Boards. HAVO requires the Electronic Message Sign Boards to enhance visitor safety, traffic management, and emergency communications.

The boards will be used to:

- Communicate volcanic eruption hazards, road closures, and evacuation instructions
- Support construction and maintenance operations by clearly notifying visitors of changing traffic patterns
- Display park-wide safety information and operational updates
- Enforce temporary MPH zones using integrated radar speed detection to improve driver compliance and reduce accident risk
- Provide high-visibility warnings in areas affected by volcanic gases, unstable terrain, or reduced visibility

Primary requirements to meet the parks minimum needs are:

- Full-matrix LED display capable of text, symbols, and variable messaging
- Remote programming capability for rapid message updates
- Integrated radar speed sensing to display real-time vehicle speeds
- Portable / deployable for temporary or emergency use
- Designed for outdoor use in hazardous and rapidly changing environments
- Compatible with traffic control, safety messaging, and emergency operation

A. MINIMUM SALIENT CHARACTERISTICS or GENERALLY ACCEPTABLE SPECIFICATIONS

Note: All products must meet the requirements set forth by the BUY AMERICAN ACT.

Overview of Required Items

Item	Item Quantity	Item Description
Electronic Message Sign Boards	[2] [Standard commercial units]	Full matrix, variable message display capable of multi-line text and basic graphics; high-resolution amber LED modules with automatic brightness control; corrosion-resistant aluminum or powder-coated steel; weatherproof enclosure rated for continuous outdoor use. Lockable control and battery box for theft protection. Cellular modem for remote programming and real-time updates. Travel Height 100"-105", deployed height 159"-165", width 95"-100", overall length 135"-165"; Affixed to a trailer for simple portable deployment for temporary or emergency use. Standard colors acceptable.

Attachment 1 – Specifications (Revised 6/13/2023)

[Item #1 Electronic Message Sign Boards]

Quantity: [2]

Summary Description: Portable, full-matrix electronic variable message sign boards for roadside deployment capable of displaying multi-line text and basic graphics for traffic control, visitor information, and emergency communications. Units shall be commercial grade, designed for continuous outdoor use, and capable of remote programming.

- Full Matrix LED Display
 - Capable of multi-line text and basic graphic symbols
 - High-resolution amber LED modules
- Automatic Visibility Control
 - Automatic brightness adjustment based on ambient light conditions
 - Legible in direct sunlight and nighttime conditions
- Outdoor Durability
 - Corrosion-resistant aluminum or powder-coated steel cabinet
 - Weatherproof enclosure rated for continuous outdoor use
- Security Features
 - Lockable control panel
 - Lockable battery box for theft protection
- Remote Communication Capability
 - Integrated cellular modem
 - Real-time remote message programming and system updates
- Dimensions
 - Acceptable ranges of dimensions:
 - Travel Height: 100”-105”
 - Deployed Height: 159”-165”
 - Width: 95”-100”
 - Overall Length: 135”-165”
- Portability
 - Commercial-grade trailer designed to support and transport electronic message sign boards for safe roadside deployment and repositioning.
 - Steel structural frame construction
 - Designed to safely support mounted sign board equipment
 - Adjustable drawbar
 - Standard hitch configuration compatible with government fleet vehicles
 - DOT-compliant lighting and reflectors
 - Highway-rated tires suitable for paved park roadways
 - Stabilizing outriggers or leveling jacks for secure deployment
 - Designed for stationary roadside operation
 - Corrosion-resistant finish
 - Mechanisms to adjust sign positions
- Standard color acceptable

Safety Features

- Required safety terms.
 - Comply with applicable provisions of the Manual on Uniform Traffic Control Devices (MUTCD) for Portable Changeable Message Signs (PCMS).

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- Trailer components shall meet U.S. DOT roadway safety requirements, including operational brake lights, turn signals, running lights, reflectors, retroreflective red/white conspicuity markings, and highway-rated tires.
- Include a minimum four-point stabilization system (outriggers or leveling jacks), functional parking brake, and wheel chocks to prevent unintended movement or tipping.
- Be rated for manufacturer-specified wind loads appropriate for roadside deployment.
- Battery and control enclosures shall be lockable and weather-sealed for theft prevention and environmental protection.
- Provide remote programming capability to minimize personnel exposure to active traffic lanes.
- All equipment shall be new and free of defects affecting safety or operability.

Warranty

- Required warranty terms.
 - Minimum Warranty Period: 5-year warranty from the date of Government acceptance. If the manufacturer's standard commercial warranty exceeds 5 years for any component, the longer term shall apply. Coverage: The warranty shall cover defects in materials and workmanship arising under normal use, including but not limited to:
 - LED module failure
 - Power system defects (solar panels, charge controllers, batteries)
 - Trailer structural defects (frame, welds, mast components)
 - Paint or corrosion issues
 - Electrical component or control system failures
 - The warranty shall include repairing or replacement of defective components at no cost to the Government. Technical support should be available during normal business hours for troubleshooting and warranty coordination.

B. OTHER REQUIREMENTS

Incidental Installation

- Incidental Installation not applicable

Documentation, Manuals, and Technical Literature

The below documentation shall be furnished:

- One (1) Operation & Maintenance Manual that includes, but is not limited to, the original equipment manufacturers (OEM) technical literature for all supplied equipment, operator/safety instructions, as-built drawings, and as-built electrical system drawings.
- Original Bill of Sale
- Manufacturer's Statement of Origin documents
- Free and clear legal title(s) to the required item(s)

Keys

- All keys associated with the delivered items shall be furnished to the Government at the time of delivery, to include: two (2) complete key sets for all doors, hatches, and ignition.

C. GOVERNMENT FURNISHED PROPERTY OR SERVICES

D. DELIVERY INFORMATION

Delivery Location

- Hawaii Volcanoes National Park, 1 Crater Rim Drive HNP, HI 96718
- Items to be delivered to the HAVO rain shed.
- Contact the designated Contracting Officer's Representative 10 business days prior to delivery to coordinate delivery logistics.

Delivery Logistics

- A park-operated forklift will NOT be available for unloading. The Contractor is responsible for providing all necessary equipment to offload delivered items. The Contractor must ensure that all personnel operating equipment for delivery are appropriately licensed and qualified, as applicable.

Delivery Term

- The Contractor shall contact the COR within 7 calendar days after contract award to coordinate delivery logistics, including exact delivery dates, locations, and any special site access requirements.
- Delivery shall be made Monday through Thursday between the hours of 6:00 AM HDT to 4:00 PM HDT.

Special Inspections / Invasive Species Inspection

- See the accompanying *HAVO Invasive Species SOPs*. All Contractor vehicles, to include subcontractor vehicles, shall be inspected for invasive species prior to entering the park. Inspections shall be coordinated with the NPS representative(s) identified by the *HAVO Invasive Species SOPs* no later than 7 days prior to the intended delivery date. Invasive species inspections may take approximately 1 hour to perform. If the delivery shipment is found to be contaminated with pests, to include little fire ants (LFAs), the shipment will be rejected. If rejected, the shipment must then be treated at the vendor's expense prior to the rescheduling of a reinspection. If no pests are found, then the material will be accepted.

E. INSPECTION AND ACCEPTANCE

Inspection

- The Contracting Officer's Representative will inspect the delivered items at the time of delivery.
- The Government reserves the right to perform inspection activities within [X] business days from the date of delivery.

Acceptance

- The Government reserves the right to reject all items that do not adhere to these specifications. Any item identified through timely inspection that does not conform to these specifications may be rejected by the Government. The Contractor shall replace all non-conforming items at no cost to the Government. The Government reserves the right to seek consideration from the Contractor for failures to satisfy the requirements established by these specifications.
- The Government reserves the right to reject defective items identified through timely inspection. Defects in materials and workmanship include, but are not limited to weld cracks, paint flaking, bubbling, fading, corrosion, and other deficiencies that may require rework or redress.

F. PAYMENT

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Payment Made through the Invoice Processing Platform (IPP)

- See the IPP clause within the solicitation or contract award for payment guidance.
- Prior to submitting an invoice to IPP, all invoices shall be submitted to the Contracting Officer's Representative or Park Point of Contact for review and, if acceptable, approval.